

MANISH RAINA

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PROFILE

AI/ML Engineer (fresher) with hands-on, end-to-end project experience across model development, backend engineering, and full-stack delivery. Comfortable building with Python, Scikit-learn, TensorFlow, and FastAPI/React, with applied work in NLP-driven matching, secure authentication systems, and clinical risk prediction. Looking to bring strong fundamentals and a builder's mindset to a production ML or AI engineering team.

EDUCATION

B.Tech, Computer Science Engineering (AI & ML)

Aug 2022 – Aug 2026

Panipat Institute of Engineering & Technology (PIET), Haryana | CGPA: 7.5

Senior Secondary (Class XII) — Kotwal National Institute

Mar 2018 – Mar 2022

TECHNICAL SKILLS

- **Languages:** Python, C, C++, JavaScript, SQL
- **ML & AI:** Scikit-learn, TensorFlow, LightGBM, NLP, Embedding-based Semantic Search, Model Evaluation (AUC, Cross-Validation, Calibration, IPCW Weighting)
- **Web & Backend:** FastAPI, React.js, REST APIs, JWT Authentication, RBAC, Middleware Validation
- **Tools & Other:** Git, Web Scraping, Power BI, Speech Recognition & Synthesis (SpeechRecognition, pyttsx3), OpenCV, dlib

EXPERIENCE

Machine Learning & Web Development Intern — EISystems Technologies

- Worked on data preprocessing and ML model development for client projects, alongside front-end feature development using React.js.

Machine Learning Research Intern — NIT Delhi

- Researched and implemented supervised machine learning algorithms using Python, Scikit-learn, Pandas, and NumPy, conducted comparative performance analysis, and prepared detailed technical documentation of experimental results.

PROJECTS

Heart Disease Survival Prediction Framework

- Built a survival prediction model using LightGBM with IPCW weighting to handle censored clinical outcome data.
- Used Scikit-learn for cross-validation and calibration, with AUC-based evaluation to assess model discrimination on a heart disease dataset. GitHub: [\[link\]](#)

AI Career Skill Intelligence Platform (Full-Stack + ML + Scraper)

- Designing and building an end-to-end career intelligence platform using FastAPI, React, and PostgreSQL.
- Implementing NLP-based resume parsing, skill extraction, and embedding-based semantic matching between job listings and candidate profiles.
- Built a web scraper for job-trend analysis and a dashboard to visualize in-demand skills. GitHub: [\[link\]](#)

Industry-Level Authentication & Access Control System

- Built a secure authentication backend with FastAPI — JWT access/refresh tokens, RBAC, password hashing, and middleware-based request validation.
- Designed protected API routes and session management for. GitHub: [\[link\]](#)

AI Voice Receptionist (Speech-to-Speech AI Assistant)

- Built a voice assistant pipeline using SpeechRecognition and PyAudio for input capture, and pyttsx3 for speech output.
- Integrated OpenCV, dlib, and face_recognition for a face-recognition layer alongside the voice interaction flow.
- Designed rule-based conversation logic for automated query handling. GitHub: [\[link\]](#)

CERTIFICATIONS & ACHIEVEMENTS

- Machine Learning with Python — Internship Certificate, EISystems Technologies
- Participant — Built a real-time video calling system for deaf & mute accessibility, HackOps, Savisjar (2024)
- Machine Learning Research Internship — NIT Delhi (2025)
- Advanced Diploma in Software Technology (12 Months) — Supertech (India) Computer Education